

第一章习题(I)解答

1. 求出以下一阶线性常微分方程解的表达式:

$$y' + q(x)y = f(x), \quad y(x_0) = y_0, \quad x \in (a, b). \quad (0.1)$$

这里 $q(x), f(x) \in C(a, b), x_0 \in (a, b)$.

解: 在(0.1)两边同乘以 $e^{\int_{x_0}^x q(t)dt}$, 可得:

$$(y(x)e^{\int_{x_0}^x q(t)dt})' = f(x)e^{\int_{x_0}^x q(t)dt}. \quad (0.2)$$

(0.2)两边从 x_0 到 x 积分, 可得

$$y(x)e^{\int_{x_0}^x q(t)dt} - y(x_0) = \int_{x_0}^x f(t)e^{\int_{x_0}^t q(s)ds} dt.$$

化简即得:

$$y(x) = e^{-\int_{x_0}^x q(t)dt} [y(x_0) + \int_{x_0}^x f(t)e^{\int_{x_0}^t q(s)ds} dt], \quad x \in (a, b). \quad (0.3)$$

此式也可以写成以下等价的形式:

$$y(x) = \frac{y(x_0) + \int_{x_0}^x f(t)e^{\int_{x_0}^t q(s)ds} dt}{e^{\int_{x_0}^x q(t)dt}}, \quad x \in (a, b). \quad (0.4)$$

2. 考虑以下一阶线性常微分方程:

$$y' + \alpha y = f(x), \quad x \in [x_0, +\infty). \quad (0.5)$$

这里 α 是非零实常数, $f(x) \in C(x_0, +\infty)$, 且积分 $\int_{x_0}^{+\infty} f(x)e^{\alpha(x-x_0)} dx$ 存在有限. 证明:

(i). 当 $\alpha > 0$ 且 $\lim_{x \rightarrow +\infty} f(x) = A$ 存在有限时. 则(0.5)的解均具有有以下极限:

$$\lim_{x \rightarrow +\infty} y(x) = \frac{Ae^{\alpha x_0}}{\alpha}.$$

(ii). 当 $\alpha < 0$, (0.5) 只有一个解具有有限极限: $\lim_{x \rightarrow +\infty} y(x)$. 并求出此解的表达式(即求出 $y(x_0) = ?$)

证明: (i). 由(0.4), 令 $q(x) = \alpha$, 则 $\int_{x_0}^t q(s)ds = \alpha(t - x_0)$, (0.5) 的满足 $y(x_0) = y_0$ 的解为

$$y(x) = \frac{y(x_0) + \int_{x_0}^x f(t)e^{\alpha(t-x_0)}dt}{e^{\alpha(x-x_0)}}. \quad (0.6)$$

(0.6) 两边令 $x \rightarrow +\infty$, 并用 *L'Hospital* 法则, 可得

$$\lim_{x \rightarrow +\infty} y(x) = \lim_{x \rightarrow +\infty} \frac{f(x)e^{\alpha(x-x_0)}}{\alpha e^{\alpha(x-x_0)}} = \frac{A}{\alpha}.$$

(ii). 当 $\alpha < 0$ 时, $\lim_{x \rightarrow +\infty} e^{\alpha(x-x_0)} = 0$. 这时(0.6) 的分子只有一个满足: $y(x_0) + \int_{x_0}^{+\infty} f(x)e^{\alpha(x-x_0)}dx = 0$, 即 $y(x_0) = -\int_{x_0}^{+\infty} f(x)e^{\alpha(x-x_0)}dx$. 这时

$$y(x) = \frac{-\int_{x_0}^{+\infty} f(t)e^{\alpha(t-x_0)}dt + \int_{x_0}^x f(t)e^{\alpha(t-x_0)}dt}{e^{\alpha(x-x_0)}} = \frac{-\int_x^{+\infty} f(t)e^{\alpha(t-x_0)}dt}{e^{\alpha(x-x_0)}}. \quad \square$$

3. 求出以下 Bernoulli 方程的解的表达式:

$$y' + p(x)y = q(x)y^n, \quad (n \neq 0, 1), \quad p(x), q(x) \in C[a, b], \quad y(x_0) = y_0, \quad x_0 \in [a, b]. \quad (0.7)$$

解: 在(0.7) 两边同乘以 $(1-n)y^{-n}(x)$ 并令 $z(x) = y^{1-n}(x)$. 则 $z' = (1-n)y^{-n}y'$, 这时(0.7)退化成为以下一阶线性常微分方程:

$$z' + (1-n)p(x)z = q(x)(1-n), \quad (0.8)$$

在(0.8)两边同乘以 $e^{(1-n)\int_{x_0}^x p(t)dt}$, 并从 x_0 到 $x \in [a, b]$ 积分, 即得

$$e^{(1-n)\int_{x_0}^x p(t)dt} z(x) = z(x_0) + (1-n) \int_{x_0}^x e^{(1-n)\int_{x_0}^t p(s)ds} q(t)dt.$$

上式两边同乘以 $e^{(n-1)\int_{x_0}^x p(t)dt}$, 即得

$$y^{1-n}(x) = z(x) = e^{(n-1)\int_{x_0}^x p(t)dt} \left[y^{1-n}(x_0) + (1-n) \int_{x_0}^x e^{(1-n)\int_{x_0}^t p(s)ds} q(t)dt \right].$$

上式两边开 $1-n$ 次方即得 $y(x)$.

4. 证明以下Bellman型不等式(I): 设 $u(x) \in C[a, b]$ 且 $u(x) \geq 0, \forall x \in [a, b]$, 若 $\alpha > 0, \beta \geq 0$ 是常数, 且 $u(x)$ 满足:

$$u(x) \leq u(a) + \int_a^x [\alpha u(t) + \beta] dt, \quad x \in [a, b]. \quad (0.9)$$

则 $u(x)$ 满足以下不等式:

$$u(x) \leq u(a)e^{\alpha(x-a)} + \frac{\beta}{\alpha} [e^{\alpha(x-a)} - 1] = \left[u(a) + \frac{\beta}{\alpha} \right] e^{\alpha(x-a)} - \frac{\beta}{\alpha}, \quad \forall x \in [a, b].$$

证明: 记 $v(x) = u(a) + \int_a^x [\alpha u(t) + \beta] dt, x \in [a, b]$, 则 $u(x) \leq v(x), \forall x \in [a, b]$ 且 $v(x) \in C^1[a, b]$, 对 $v(x)$ 求导, 并利用 $\alpha > 0$ 可得

$$v'(x) = \alpha u(x) + \beta \leq \alpha v(x) + \beta.$$

即

$$v'(x) - \alpha v(x) \leq \beta.$$

上式两边同乘以 $e^{-\alpha x}$, 并从 a 到 $x \in [a, b]$ 积分, 可得:

$$v(x)e^{-\alpha x} - v(a)e^{-\alpha a} \leq \beta \int_a^x e^{-\alpha t} dt = \frac{\beta}{\alpha} (e^{-\alpha a} - e^{-\alpha x}).$$

上式两边同乘以 $e^{\alpha x}$ 并注意到 $v(a) = u(a)$ 可得

$$v(x) \leq v(a)e^{\alpha(x-a)} + \frac{\beta}{\alpha} [e^{\alpha(x-a)} - 1] = u(a)e^{\alpha(x-a)} + \frac{\beta}{\alpha} [e^{\alpha(x-a)} - 1].$$

又因 $u(x) \leq v(x), \forall x \in [a, b]$, 即得

$$u(x) \leq u(a)e^{\alpha(x-a)} + \frac{\beta}{\alpha} [e^{\alpha(x-a)} - 1] = \left[u(a) + \frac{\beta}{\alpha} \right] e^{\alpha(x-a)} - \frac{\beta}{\alpha}. \quad \square$$

5. 证明以下Bellman型不等式(II): 设 $u(x), \alpha(x), \beta(x), p(x) \in C[a, b]$ 且 $u(x), \alpha(x), \beta(x), p(x) \geq 0, \forall x \in [a, b]$.

求证(A). 若

$$u(x) \leq \alpha(x) + \int_a^x \beta(t)u(t)dt, \quad \forall x \in [a, b], \quad (0.10)$$

则

$$u(x) \leq \alpha(x) + \int_a^x \alpha(t)\beta(t)e^{\int_t^x \beta(s)ds}dt, \quad \forall x \in [a, b].$$

(B). 在(A)的假设下, 若 $\alpha(x)$ 还是单调不减函数, 则

$$u(x) \leq \alpha(x)e^{\int_t^x \beta(s)ds}, \quad \forall x \in [a, b].$$

(C). 在(A)的假设下, 若

$$u(x) \leq u(a) + \int_a^x [p(t)u(t) + \beta(t)] dt, \quad \forall x \in [a, b].$$

证明:

$$u(x) \leq \frac{u(a) + \int_a^x \beta(t)e^{-\int_a^t p(s)ds} dt}{e^{-\int_a^x p(t)dt}}, \quad \forall x \in [a, b].$$

证明(A): 记

$$v(x) = \int_a^x \beta(t)u(t)dt, \quad x \in [a, b],$$

则 $v(a) = 0$, $u(x) \leq \alpha(x) + v(x)$, $\forall x \in [a, b]$ 且 $v(x) \in C^1[a, b]$, 对 $v(x)$ 求导, 可得

$$v'(x) = \beta(x)u(x) \leq \beta(x)[\alpha(x) + v(x)] = \beta(x)v(x) + \alpha(x)\beta(x),$$

即

$$v'(x) - \beta(x)v(x) \leq \alpha(x)\beta(x).$$

上式两边同乘以 $e^{-\int_a^x \beta(t)dt}$, 并从 a 到 $x \in [a, b]$ 积分, 并注意到 $v(a) = 0$ 可得:

$$v(x)e^{-\int_a^x \beta(t)dt} \leq \int_a^x \alpha(t)\beta(t)e^{-\int_a^t \beta(s)ds} dt.$$

上式两边同乘以 $e^{\int_a^x \beta(t)dt}$ 可得

$$v(x) \leq \int_a^x \alpha(t)\beta(t) \left(e^{\int_a^x \beta(t)dt - \int_a^t \beta(s)ds} \right) dt = \int_a^x \alpha(t)\beta(t)e^{\int_t^x \beta(s)ds} dt,$$

从而有

$$u(x) \leq \alpha(x) + v(x) \leq \alpha(x) + \int_a^x \alpha(t)\beta(t)e^{\int_t^x \beta(s)ds} dt.$$

证明(B): 由(A)的证明可得, 对 $x \geq a$,

$$\begin{aligned} u(x) &\leq \alpha(x) + \int_a^x \alpha(t) \beta(t) e^{\int_t^x \beta(s) ds} dt \\ &\leq \alpha(x) + \alpha(x) \int_a^x \beta(t) e^{\int_t^x \beta(s) ds} dt \\ &= \alpha(x) \left[1 + \int_a^x \beta(t) e^{\int_t^x \beta(s) ds} dt \right] \\ &= \alpha(x) \left[1 - \int_a^x \frac{d}{dt} \left(e^{\int_t^x \beta(s) ds} \right) dt \right] \\ &= \alpha(x) \left[1 - e^{\int_t^x \beta(s) ds} \Big|_a^x \right] \\ &= \alpha(x) e^{\int_a^x \beta(t) dt}. \end{aligned}$$

证明(C): 令 $v(x) = u(a) + \int_a^x [p(t)u(t) + \beta(t)] dt$, $\forall x \in [a, b]$, 则 $v(a) = u(a)$, $u(x) \leq v(x)$, $\forall x \in [a, b]$.

$$v'(x) = p(x)u(x) + \beta(x) \leq p(x)v(x) + \beta(x).$$

等式两边同乘以 $e^{-\int_a^x p(t)dt}$, 可得

$$\left(v(x)e^{-\int_a^x p(t)dt}\right)' \leq \beta(x)e^{-\int_a^x p(t)dt},$$

等式两边从 a 到 $a \leq x \leq b$ 积分, 并由 $v(a) = u(a)$, 可得

$$\int_a^x \left(v(x) e^{-\int_a^x p(t) dt} \right)' dx = v(x) e^{-\int_a^x p(t) dt} - v(a) = v(x) e^{-\int_a^x p(t) dt} - u(a) \leq \int_a^x \beta(t) e^{-\int_a^t p(s) ds} dt,$$

可得

$$u(x) \leq v(x) \leq \frac{u(a) + \int_a^x \beta(t) e^{-\int_a^t p(s) ds} dt}{e^{-\int_a^x p(t) dt}}, \quad \forall x \in [a, b]. \square$$

6.考虑以下 n 维一阶常微分方程系统:

[illegible]

这里 $x \in [a, b]$, $f_k(x; y_1, y_2, \dots, y_n)$, $k = 1, 2, \dots, n$ 是各变量的连续函数, 且存在连续非负函数 $L(x) \in C[a, b]$, 对任意 $Y = (y_1, y_1, \dots, y_n)$, $Z = (z_1, z_2, \dots, z_n) \in R^n$, $x \in [a, b]$, 以下不等

式成立:

$$|f_k(x; z_1, z_2, \dots, z_n) - f_k(x; y_1, y_2, \dots, y_n)| \leq L(x) \sum_{j=1}^n |z_j - y_j|, \quad k = 1, 2, \dots, n. \quad (0.12)$$

证明: 若(0.11)有满足初始条件 $y_k(x_0; y_0^1, y_0^2, \dots, y_0^n) = y_0^k, k = 1, 2, \dots, n$, 的解 $Y(x) = (y_1(x; y_0^1, y_0^2, \dots, y_0^n), y_2(x; y_0^1, y_0^2, \dots, y_0^n), \dots, y_n(x; y_0^1, y_0^2, \dots, y_0^n))$. 则解是唯一的.

证明: 若(0.11)有两个解 $Y(x), Z(x)$ 满足相同的初始条件 $Y(x_0) = Z(x_0) = (y_0^1, y_0^2, \dots, y_0^n)$.

记

$$V(x) = \|Y - Z\|^2 = \sum_{k=1}^n (y_k(x; y_0^1, y_0^2, \dots, y_0^n) - z_k(x; y_0^1, y_0^2, \dots, y_0^n))^2 =: \sum_{k=1}^n (y_k(x) - z_k(x))^2.$$

则 $V \in C^1[a, b]$, 且由(0.12), 可得

$$\begin{aligned} V'(x) &= 2 \sum_{k=1}^n (y_k(x) - z_k(x))(y'_k(x) - z'_k(x)) \\ &= 2 \sum_{k=1}^n (y_k(x) - z_k(x))(f_k(x; y_1, y_2, \dots, y_n) - f_k(x; z_1, z_2, \dots, z_n)) \\ &\leq 2 \sum_{k=1}^n |y_k(x) - z_k(x)| L(x) \sum_{k=1}^n |y_k(x) - z_k(x)| \\ &= 2L(x) (\sum_{k=1}^n |y_k(x) - z_k(x)|)^2 \\ &\leq 2nL(x) \sum_{k=1}^n (y_k(x) - z_k(x))^2 \\ &= 2nL(x)V(x). \end{aligned}$$

这里用到了Cauchy不等式:

$$\left(\sum_{k=1}^n a_k b_k \right)^2 \leq \left(\sum_{k=1}^n a_k^2 \right) \cdot \left(\sum_{k=1}^n b_k^2 \right), \quad a_k = |y_k(x) - z_k(x)|, \quad b_k = 1, \quad k = 1, 2, \dots, n.$$

即 $V'(x) \leq 2nL(x)V(x), \forall a \leq x_0 \leq x \leq b$. 由此可得 $\left(V(x)e^{-2n \int_{x_0}^x L(t)dt} \right)' \leq 0$. 对上式从 x_0 到 x 积分, 并注意到 $V(x_0) = 0$, 可得不等式:

$$0 \leq V(x)e^{-2n \int_{x_0}^x L(t)dt} - V(x_0) = V(x)e^{-2n \int_{x_0}^x L(t)dt} \leq 0,$$

这等价于

$$0 \leq V(x) \leq V(x_0)e^{2n \int_{x_0}^x L(t)dt} = 0, \quad V(x) \equiv 0, \quad \forall x \in [x_0, b] \subset [a, b].$$

这就证明了 $Y(x) = Z(x), \forall x_0 \leq x \leq b$.

推论：若 $f_k(x; y_1, y_2, \dots, y_n), k = 1, 2, \dots, n$, 对各个变量均有连续偏导数，则在有限区域 D 内， $\max |\frac{\partial f_k}{\partial y_j}|$ 有界，即存在常数 $L > 0$ 使得

$$\max_{1 \leq i, j \leq n} \left| \frac{\partial f_k}{\partial y_j} \right| \leq L.$$

这时可用常数 L 代替连续函数 $L(x)$, 原命题仍正确 (Lipschitz 条件)，即若 (0.11) 有满足初始条件 $y_k(x_0; y_0^1, y_0^2, \dots, y_0^n) = y_0^k, k = 1, 2, \dots, n$, 的解

$$Y(x) = (y_1(x; y_0^1, y_0^2, \dots, y_0^n), y_2(x; y_0^1, y_0^2, \dots, y_0^n), \dots, y_n(x; y_0^1, y_0^2, \dots, y_0^n)).$$

则解是唯一的.

证明：若 (0.11) 有两个解 $Y(x), Z(x)$ 满足相同的初始条件 $Y(x_0) = Z(x_0) = (y_0^1, y_0^2, \dots, y_0^n)$.

记

$$Y = (y_1, y_2, \dots, y_n), \quad Z = (z_1, z_2, \dots, z_n) \in D$$

则有

$$\begin{aligned} & |f_k(x; z_1, z_2, \dots, z_n) - f_k(x; y_1, y_2, \dots, y_n)| \\ & \leq |f_k(x; z_1, z_2, \dots, z_n) - f_k(x; y_1, z_2, \dots, z_n)| \\ & \quad + |f_k(x; y_1, z_2, \dots, z_n) - f_k(x; y_1, y_2, z_3, \dots, z_n)| \\ & \quad + |f_k(x; y_1, y_2, z_3, \dots, z_n) - f_k(x; y_1, y_2, y_3, z_4, \dots, z_n)| \\ & \quad + |f_k(x; y_1, y_2, y_3, \dots, y_{n-1}, z_n) - f_k(x; y_1, y_2, y_3, \dots, y_{n-1}, y_n)| \\ & \leq \max \left| \frac{\partial f_k}{\partial y_1} \right| |z_1 - y_1| + \max \left| \frac{\partial f_k}{\partial y_2} \right| |z_2 - y_2| + \dots + \max \left| \frac{\partial f_k}{\partial y_n} \right| |z_n - y_n| \\ & \leq L \cdot [\sum_{k=1}^n |z_k - y_k|], \quad k = 1, 2, \dots, n. \end{aligned} \tag{0.13}$$

记

$$V(x) = \|Y - Z\|^2 = \sum_{k=1}^n (y_k(x; y_0^1, y_0^2, \dots, y_0^n) - z_k(x; y_0^1, y_0^2, \dots, y_0^n))^2 =: \sum_{k=1}^n (y_k(x) - z_k(x))^2.$$

则 $V \in C^1[a, b]$, 且由(0.13), 可得

$$\begin{aligned}
V'(x) &= 2 \sum_{k=1}^n (y_k(x) - z_k(x))(y'_k(x) - z'_k(x)) \\
&= 2 \sum_{k=1}^n (y_k(x) - z_k(x))(f_k(x; y_1, y_2, \dots, y_n) - f_k(x; z_1, z_2, \dots, z_n)) \\
&\leq 2 \sum_{k=1}^n |y_k(x) - z_k(x)| L \sum_{k=1}^n |y_k(x) - z_k(x)| \\
&= 2L (\sum_{k=1}^n |y_k(x) - z_k(x)|)^2 \\
&\leq 2nL \sum_{k=1}^n (y_k(x) - z_k(x))^2 \\
&= 2nLV(x).
\end{aligned}$$

这里用到了Cauchy不等式:

$$\left(\sum_{k=1}^n a_k b_k \right)^2 \leq \left(\sum_{k=1}^n a_k^2 \right) \cdot \left(\sum_{k=1}^n b_k^2 \right), \quad a_k = |y_k(x) - z_k(x)|, \quad b_k = 1, \quad k = 1, 2, \dots, n.$$

即 $V'(x) \leq 2nLV(x), \forall a \leq x_0 \leq x \leq b$. 由此可得 $(V(x)e^{-2nLx})' \leq 0$. 对上式从 x_0 到 x 积分, 并注意 $V(x_0) = 0$, 可得不等式:

$$0 \leq V(x) \leq V(x_0)e^{2nL(x-x_0)} = 0, \quad V(x) \equiv 0, \quad \forall x \in [x_0, b] \subset [a, b].$$

这就证明了 $Y(x) \equiv Z(x), \forall x_0 \leq x \leq b$. $\square$

7. 设 $y(x)$ 在 $I = [0, +\infty)$ 上连续可微, 且存在常数 $\alpha > 0$, 满足

$$\lim_{x \rightarrow +\infty} (y'(x) + \alpha y(x)) = 0.$$

证明: $\lim_{x \rightarrow +\infty} y(x) = 0$.

证明: 记 $y'(x) + \alpha y(x) = f(x)$, 等式两边同乘以 $e^{\alpha x}$, 并从 $x = 0$ 到 $x > 0$ 积分, 可得 $y(x)e^{\alpha x} = y(0) + \int_0^x e^{\alpha t} f(t) dt$, 即

$$y(x) = \frac{y(0) + \int_0^x e^{\alpha t} f(t) dt}{e^{\alpha x}}.$$

由L'Hospital 法则, 可得

$$\lim_{x \rightarrow +\infty} y(x) = \lim_{x \rightarrow +\infty} \frac{y(0) + \int_0^x e^{\alpha t} f(t) dt}{e^{\alpha x}} = \lim_{x \rightarrow +\infty} \frac{e^{\alpha x} f(x)}{\alpha e^{\alpha x}} = \lim_{x \rightarrow +\infty} \frac{f(x)}{\alpha} = 0. \quad \square$$

8. 证明: 若 $y(x)$ 是方程

$$y' = f(x) \sin y$$

的满足条件 $y(0) = 0$ 的解, 则 $y(x) \equiv 0$, 这里 $f(x) \in C(-\infty, +\infty)$.

证明: 设 $y(x)$ 是此方程满足初始条件 $y(0) = y_0$ 的解. 令 $v(x) = y^2(x)$, 则

$$v'(x) = 2y(x)y'(x) = 2f(x)y(x) \sin y(x) \leq 2|f(x)|y^2(x) = 2|f(x)|v(x). \quad (0.14)$$

这里用到了不等式 $|\sin y| \leq |y|$. (0.14)等价于以下不等式: $v'(x) - 2|f(x)|v(x) \leq 0$. 在以上不等式两边同乘以 $e^{-2 \int_0^x |f(t)|dt}$, 并从 $x = 0$ 到 $x > 0$ 积分, 可得

$$0 \leq v(x)e^{-2 \int_0^x |f(t)|dt} \leq v(0) = y^2(0) = y_0^2.$$

特别的, 当 $y(0) = y_0 = 0$ 时, 有 $y^2(x) = v(x) \equiv 0$. $\square$

9. 设 $y_1(x)$, $y_2(x)$ 是方程(0.1)的两个互异解. 证明对于(0.1)的任一解 $y(x)$, 成立恒等式

$$\frac{y(x) - y_1(x)}{y_2(x) - y_1(x)} = C, \quad (0.15)$$

这里 C 是某个常数.

证明: 因(0.1)的任一解可写成

$$y(x) = \frac{y(x_0) + \int_{x_0}^x f(t)e^{\int_{x_0}^t q(s)ds} dt}{e^{\int_{x_0}^x q(t)dt}}, \quad x \in (a, b). \quad (0.16)$$

在(0.15)中分别令 $y(x) = y_1(x)$, $y(x) = y_2(x)$, $y(x) = y(x)$, $y_1(x_0) = c_1$, $y_2(x_0) = c_2$, $y(x_0) = c$.

则由 $y_1(x) \neq y_2(x)$, 知 $c_1 \neq c_2$, 可得

$$\frac{y(x) - y_1(x)}{y_2(x) - y_1(x)} = \frac{y(x_0) - y_1(x_0)}{y_2(x_0) - y_1(x_0)} = \frac{c - c_1}{c_2 - c_1} = C. \square$$

附录: Lagarange Identity

$$\left(\sum_{k=1}^n a_k b_k \right)^2 + \sum_{1 \leq k < j \leq n} (a_k b_j - a_j b_k)^2 = \left(\sum_{k=1}^n a_k^2 \right) \left(\sum_{k=1}^n b_k^2 \right).$$

$n = 3$ 时, 有下列向量表达式

$$(\mathbf{A} \cdot \mathbf{B})^2 + (\mathbf{A} \times \mathbf{B})^2 = \|\mathbf{A}\|^2 \|\mathbf{B}\|^2.$$

即当 $\mathbf{A} = (a_x, a_y, a_z)$, $\mathbf{B} = (b_x, b_y, b_z)$ 时, 有恒等式:

$$(a_y b_z - a_z b_y)^2 + (a_z b_x - a_x b_z)^2 + (a_x b_y - a_y b_x)^2 = (a_x^2 + a_y^2 + a_z^2)(b_x^2 + b_y^2 + b_z^2) - (a_x b_x + a_y b_y + a_z b_z)^2.$$

由LagarangeIdentity 立即可得以下Cauchy不等式：若 $\{a_k\}, \{b_k\}$ 是实数数列，则以下Cauchy不等式成立：

$$\left(\sum_{k=1}^n a_k b_k\right)^2 \leq \left(\sum_{k=1}^n a_k^2\right) \left(\sum_{k=1}^n b_k^2\right),$$

等式成立当且仅当

$$\frac{a_1}{b_1} = \frac{a_2}{b_2} = \cdots = \frac{a_n}{b_n}.$$

更一般的，证明： $y' = f(x) \sin y$, $y(0) = y_0$ 的解唯一。

证明：设有两个解 $y(x)$, $z(x)$ 分别满足以上初值问题： $y' = f(x) \sin y$, $z' = f(x) \sin z$, $y(0) = z(0) = y_0$. 令 $V(x) = (y(x) - z(x))^2$. 则 $V(x) \geq 0, \forall x \geq 0$, 再用到Lagrange中值定理, 可得

$$\begin{aligned} V'(x) &= 2(y(x) - z(x))(y'(x) - z'(x)) \\ &= 2(y(x) - z(x))f(x)(\sin y(x) - \sin z(x)) \\ &= 2(y(x) - z(x))f(x)(y(x) - z(x)) \cos \zeta(x), \quad \zeta(x) \in (y(x), z(x)) \text{ 或 } \zeta(x) \in (z(x), y(x)) \\ &\leq 2(y(x) - z(x))^2 |f(x)| \\ &= 2|f(x)|V(x), \end{aligned}$$

由此可得

$$V'(x) - 2|f(x)|V(x) \leq 0,$$

由此可得

$$\left(V(x)e^{-2\int_0^x |f(t)|dt}\right)' \leq 0,$$

以上不等式从0到 $x > 0$ 积分，可得

$$0 \leq V(x)e^{-2\int_0^x |f(t)|dt} \leq V(0) = 0, \quad \forall x \geq 0.$$

即

$$V(x) \equiv 0, \quad \forall x \geq 0.$$

因而有

$$y(x) \equiv z(x), \quad \forall x \geq 0.$$

特别的，若令 $z(x) = z(0) = 0$ ，可得 $y(x) = z(x) = z(0) = 0$. $\square$